

Dustin Lionel

Backend Engineer

+62 812-7454-0598 | dustinlional.pro@gmail.com | [LinkedIn](#) | [Portfolio](#)

PROFILE

Backend Engineer with 1 year of production experience: 480+ merged pull requests on HR platforms serving tens of thousands of employees, and database optimizations of up to 99.8%. Strong in reliable business workflows, secure access control, query performance, and AI integration. Informatics Engineering graduate (GPA 3.94), aiming to build secure, high-impact technology for digital banking in Indonesia.

WORK EXPERIENCE

Backend Engineer

October 2025 – Present

PT. Rakamin Kolektif Madani

- **480+ merged pull requests in 12 months** across 4 HR technology products, working with Product Managers, Designers, QA, and Frontend Engineers.

Talent Management System

- **~39,000 employees** served by 3 core modules I engineered (development planning, career path recommendation, succession planning), including multi-stage approvals and committee voting.
- **41 min → 3.7 s (99.8% faster)** dashboard aggregation query, by finding the bottleneck through query-plan analysis and removing an unnecessary join, with identical results.
- **3–4 min → seconds** page-load query, by adding a targeted index on a 630K-row table with zero write downtime.
- **500K+ row-by-row inserts** replaced with a single set-based write, stabilizing a nightly reporting job that had grown to 11.5 minutes and kept failing.
- Rule-based career recommendation engine that matches employees to roles by competency, with versioned and auditable decision traces.
- Role-based access control across a multi-level organizational hierarchy, so each manager can access only their own reporting line.

AI-Powered Mentoring & Performance Platform

- **4-stage AI pipeline** for mentoring sessions (recording → transcription → summary and action items → translation), running on background job queues.
- **1-hour sync delay eliminated** by adding timeout-aware retries to a two-way calendar sync (every 15 minutes, 200 sessions per batch), alongside Single Sign-On (SSO) integration.
- **76% faster test suite (100 s → 24 s)** by cutting fixture setup from 410 inserts to 5.
- Idempotent background jobs, lock-contention fixes, and automatic stuck-session recovery to prevent duplicate processing and stalled sessions.

Other Projects

- **45-minute AI interviews ending at ~15 minutes:** fixed the interview engine's coverage logic and made its tuning configurable at runtime without redeployment.
- **401 errors after Google OAuth sign-up:** traced to a password-hashing race condition and fixed it for new users.
- AI-assisted bug-fixing workflow that turns a bug reported in team chat into a proposed fix for engineers to review.

Backend Developer

June 2023 – July 2023

Freelance

- Built and deployed the official website of **Perhimpunan Dokter Keluarga Indonesia (PDKI)** for thousands of medical professionals, using Next.js, Express.js, and MySQL.

EDUCATION

Bachelor of Informatics Engineering | GPA 3.94 / 4.00

2021 – 2025

Mikroskil University

ORGANIZATION

Vice President, Mikroskil Programming Club

October 2023 – October 2024

Mikroskil University

- **4 national and international competitions** entered: GEMASTIK, COMPFEST, ARKAVIDIA, and ICPC.
- Led member recruitment, taught C++ and competitive programming, and chaired the Ideafuse 2023 competition committee.

PROJECTS

EduCraft – AI Question Extraction App (Flutter, Flask, GPT-4o)

December 2024

Mikroskil University lecturer research

- Led development of a mobile app that extracts practice questions from documents with GPT-4o, with a Flask backend on Google Cloud.

Planitorium – Plant Disease Detection App (Kotlin)

December 2024

Bangkit Academy 2024 Batch 2 Capstone

- Built an Android app with machine-learning plant disease detection, integrated with cloud APIs.

SKILLS

- **Languages:** JavaScript, TypeScript, Java, SQL, Ruby, Python, Kotlin, C++
- **Backend:** Node.js, Express.js, Spring Boot, Ruby on Rails, Sequelize, BullMQ, REST API design, JWT / SSO (OAuth 2.0)
- **Databases:** PostgreSQL (query optimization, indexing, transactions, locking), MySQL, Redis
- **AI Integration:** Google Gemini, OpenAI GPT, Whisper, prompt engineering
- **Tools:** Git, GitHub Actions (CI/CD), Docker, Jest, RSpec, React, Next.js
- **Certifications:** Red Hat Enterprise Linux System Administration I; Android Application Development Intermediate (Dicoding); Introduction to Cloud Development (IBM)
- **Languages Spoken:** Bahasa Indonesia, English